

Course Code	21EIC552J	Course Name	INDUSTRIAL PROCESS AUTOMATION	Course Category		Professional Core	L	T	P	C
							3	0	2	4

Pre-requisite Courses	Nil	Co-requisite Courses	Nil	Progressive Courses	Nil
Course Offering Department	Electronics and Instrumentation Engineering	Data Book / Codes / Standards	Nil		

Course Learning Rationale (CLR):	The purpose of learning this course is to:
CLR-1:	Understand the components of automation system structure and different automation strategies
CLR-2:	Introduce the PLC logic program for control application
CLR-3:	Provide basic knowledge in SCADA and DCS in the field of automation

Course Outcomes (CO):	At the end of this course, learners will be able to:	Programme Outcomes (PO)		
		1	2	3
CO-1:	Select the appropriate I/O subsystem in automation system structure for process automation	2		2
CO-2:	Develop the logic program for process control application	3	2	
CO-3:	Apply the most appropriate automation technologies for a given application	3	2	

Module-1 - Automation System Structure	15 Hour
Overview of Industrial Automation, Physical Processes, Localized and Distributed Processes, Process Behavior, Process Signals, Automation Steps, PLC control, panel wiring, Data Analysis, Monitoring, and Decision Making, Benefits of Automation, Subsystems, Input Instrumentation Subsystem, Output Instrumentation Subsystem, Human Interface Subsystem, Control Subsystem, Control Execution, Instrumentation Subsystem Structure, Signal Interfacing Standards, Input Data Reliability Enhancement, Isolation and Protection, Human Interface Subsystem, Operator Panel, Structure, Interfacing, Human Interface, Discrete Process Control, Continuous Process Control, Hybrid Process Control, Advanced Strategies, Continuous Control, Feed-Forward Control, Cascade Control, Ratio Control Multiple-Step Control, Programmable Control Subsystems, Soft-wired Technology, Automation Strategy Implementation, Sequential Control with Interlocks	
Module-2 - PLC Programming and Troubleshooting	30 Hour

PLC size and application, Functional Components of a PLC, Internal Architecture, Input Devices, Output Devices, Ladder Programming, Cascading timer, counters, Cascading Counter, Combining counter and timer functions, Data compare, Math instructions, Function Block Programming, Instruction Lists, Sequential Function Charts, Structured Text Programming Method, Commissioning, Troubleshooting PLC hardware

1. Study of PLC Market survey on different PLC's, and its comparison
2. Development of control logic for filling and draining of liquid in a single tank
3. Development of control logic for Material handling
4. Development of control logic for automatic Bottle filling process
5. Development of control logic for Temperature control
6. Development of control logic for Flow control
7. Development of control logic for Lift control
8. Development of control logic for Car parking
9. Study the working of DCV's Development of control logic for Stamping machine control

Module-3 - SCADA and DCS

30 Hour

SCADA basics, Elements of SCADA, Functionality of SCADA, History of SCADA, Analog signals measurement, Control techniques, Discrete signals measurement, Control techniques, Remote terminal unit, Analog and Discrete control, Monitoring signals, Master terminal unit, Communication system components, Field/RTU communication, Communication Topology, RTU/MTU communication, System components, Monitoring alarms, Status points, Control interfacing, Parallel operator interface

Evolution of Distributed Control System, DCS Architecture, Local control unit, Architectural parameters, Operator interface, Operator Interface Requirements, Operator input-output devices, Low-level Operator Interface, Operator displays, Engineering interface- Introduction, Low-level engineering interface, High-level engineering interface, DCS Application in Power plant, Automation strategy, DCS Application in cement plant, System architecture, DCS Application in steel plant, System architecture

1. SCADA Development for the level process control training plant
2. SCADA Development for the flow process control training plant
3. SCADA Development for the temperature process control training plant
4. DCS control panel wiring diagram and creating control panel layout Interfacing level transmitter to a DCS
5. On line monitoring and control of level process using DCS
6. A mini project in process automation

Learning Resources	1. KLS Sharma "Overview of Industrial Process Automation" Elsevier, Second edition, 2017.	4. Stuart Boyer A, "SCADA : Supervisory control and data Acquisition", Fourth Edition, ISA-The Instrumentation, Systems, and Automation Society, 2010
	2. Frank D. Petruzella, "Programmable Logic Controller", Tata McGraw Hill 5th Edition, 2017.	5. IDC Technologies, "Practical Distributed Control Systems (DCS) for Engineers and Technicians" 2012.
	3. B.R. Mehta, Y. J. Reddy "Industrial Process Automation Systems Design and Implementation" Elsevier, 2015 "	6. NPTEL Video Lecture series on "Industrial Automation and Control "by Prof. S. Mukhopadhyay, IIT Kharagpur

Learning Assessment			
	Bloom's	Continuous Learning Assessment (CLA)	Summative

	Level of Thinking	Formative CLA-1 Average of unit test (45%)		Life-Long Learning CLA-2 (15%)		Final Examination (40% weightage)	
		Theory	Practice	Theory	Practice	Theory	Practice
Level 1	Remember	20%	-	-	20%	20%	-
Level 2	Understand	20%	-	-	20%	20%	-
Level 3	Apply	30%	-	-	30%	30%	-
Level 4	Analyze	30%	-	-	30%	30%	-
Level 5	Evaluate	-	-	-	-	-	-
Level 6	Create	-	-	-	-	-	-
	Total	100 %		100 %		100 %	

Course Designers		
Experts from Industry	Experts from Higher Technical Institutions	Internal Experts
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